

Exercises for Process Mining

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Notes Repository

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1 Process Discovery

1.1 Alpha Miner

1.1.1 Exercise 1

Given the following event log:

$$L_1 = [\langle a, b, c, d \rangle^3, \langle a, c, b, d \rangle^2, \langle a, e, d \rangle^1]$$

build the process model using the Alpha Miner algorithm.

Solution

1.

1.1.2 Exercise 2

Given the following event log:

$$L = [\langle a, b, c, d, e, f, b, d, c, e, g \rangle, \langle a, b, d, c, e, g \rangle^2, \langle a, b, c, d, e, f, b, c, d, e, f, b, d, c, e, g \rangle]$$

Use the Alpha Miner algorithm to discover a Petri net from the event log.

Solution:

1. First, we need to compute the footprint matrix:

	A	B	C	D	E	F	G
A	#	→	#	#	#	#	#
B		#	→	→	#	←	#
C			#		→	#	#
D				#	→	#	#
E					#	→	→
F						#	#
G							#

Table 1: Footprint Matrix for the Event Log L

2. Next, we identify the sets of all transitions T_L , the set of start activities T_I and the set of end activities T_O :

- $T_L = \{a, b, c, d, e, f, g\}$
- $T_I = \{a\}$
- $T_O = \{g\}$

3. Then, we identify the pairs of sets of transitions that satisfy the Alpha Miner conditions. To find the first pairs, we look for causality relations in the footprint matrix:

$$\{(\{a\}, \{b\}), (\{b\}, \{c\}), (\{b\}, \{d\}), (\{f\}, \{b\}), (\{c\}, \{e\}), (\{d\}, \{e\}), (\{e\}, \{f\}), (\{e\}, \{g\})\}$$

4. We can then group these pairs into maximal sets:

$$\{(\{a, f\}, \{b\}), (\{e\}, \{f, g\}), \}$$

And we can remove the non-maximal sets that composed them:

$$\{(\{a\}, \{b\}), (\{f\}, \{b\}), (\{e\}, \{f\}), (\{e\}, \{g\})\}$$

5. Finally, we can construct the Workflow net, using one place for each of the identified pairs, and connecting them according to the causality relations:

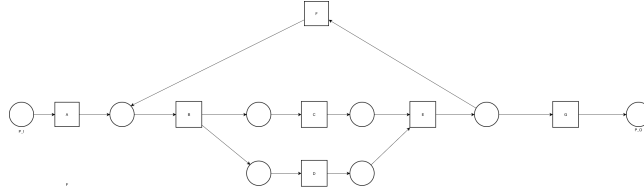


Figure 1: Petri Net Discovered using the Alpha Miner

1.1.3 Exercise 3

Given the following event log:

$$L_4 = [\langle a, c, d \rangle^4 5, \langle b, c, d \rangle^4 2, \langle a, c, e \rangle^3 8, \langle b, c, e \rangle^2 2]$$

build the process model using the Alpha Miner algorithm.

Solution

- 1.